

**Generalized Jeffreys's approximate objective Bayes factor:
Model-selection consistency, finite-sample accuracy,
and detecting the type I errors in 71,126 clinical trial findings**

<https://osf.io/qepri/>

osfstorage-archive

Figure 1

Figure 1.png

Figure 1.R

Figure 2

Figure 2.png

Figure 2.R

Figure S1.png

Figure S1.R

Figure 3

Figure 3.png

Figure 3.R

Figure S2-4.R

Figure S2.png

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setup.RData

Figure 4

Figure 4.png

Figure 4.R

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Figure S6.R

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anovaSim.Rmd

chisqSim.html

chisqSim.Rmd

coxphSim.html

coxphSim.Rmd

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anovaSim.RData
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Supplementary Information

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regularity_conditions.html
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Figure 1

Sampling distributions of the regularity condition **R2** under the alternative hypothesis.

`regularity_conditions.RData` must be loaded. The **R** scripts generate Figure 1, as presented in the manuscript and appendix.

Figure 2

Comparison of the natural logarithmic $eJA B_{10}$ or BIC approximation method with alternative Bayes-factor computations at a fixed sample size. `./Simulations/Data/* .RData` must be loaded. The **R** scripts generate Figure 2 and Figure S1 (Figure 5), as presented in the manuscript and appendix.

Figure 3

Evidence in 71,126 results and 41 sub-MeSH conditions. `setup.RData` contains saved data and functions. The **R** scripts generate Figure 3, Figure S2 (Figure 6), Figure S3 (Figure 7), and Figure S4 (Figure 8), as presented in the manuscript and appendix.

Figure 4

Measures of evidence by clinical trial phase. `setup.RData` must be loaded. The **R** scripts generate Figure 4, Figure S5 (Figure 9), and Figure S6 (Figure 10), as presented in the manuscript and appendix.

README.txt

Guidelines for using the provided HTML files.

Simulations

Simulation studies include: two-sample t -test (`ttestSim`), simple linear regression (`lmSim`), simple (binary) logistic regression (`glmSim`), one-way analysis of variance (`anovaSim`), one-way repeated-measures ANOVA (`rmanovaSim`), chi-squared test for independence (`chisqSim`), and Cox proportional-hazards regression (`coxphSim`).

We have considered several Bayes factor methods, including: the Markov chain Monte Carlo-based Bayes factor (i.e., the Savage–Dickey density ratio); the Jeffreys–Zellner–Siow and Dirichlet Bayes factors (from the ‘*BayesFactor*’ **R** package); the BIC approximation to the Bayes factor and its variations; Jeffreys’s approximate objective Bayes factor (JAB) and *eJAB*; Wagenmakers’s $3p\sqrt{n}$ rule; the Pearson Bayes factor; and the test-statistic Bayes factor.

*.RData contains saved data, variables, and functions. Each **R** Markdown renders to its respective HTML webpage, detailing the background, methodology, simulation settings, and graphic results. In particular:

ttestSim_totFALSE.RData uses the harmonic mean of the group sizes as the effective sample size, while ttestSim_totTRUE.RData uses the total number of observations.

anovaSim.RData tests three groups; anovaSim_a5.RData tests five groups; and anovaSim_2x2.RData tests a 2×2 factorial design.

rmanovaSim_rho0.9.RData and rmanovaSim_rho0.2.RData test cases with intraclass correlations of .9 (high) and .2 (low), respectively.

chisqSim_jointMulti_*.RData and chisqSim_indepMulti_*.RData test multinomial and product-multinomial designs, respectively. $R \times C$ denotes the number of rows and columns in a contingency table.

Supplementary Information - csv

CTE05.csv lists candidate type I errors identified at a significance level of 0.05 with $eJAB_{01} > 1$. It includes the data used to compute *eJAB* (statistical test, *p*-value, sample size, and parameter dimension), along with trial- and finding-specific information such as trial title, trial phase, outcome title, outcome description, comparison group description, and trial condition MeSH ID(s).

CTE01.csv lists candidate type I errors identified at a significance level of 0.01 with $eJAB_{01} > 1$. As with CTE05.csv, it includes the data used to compute *eJAB* and detailed information related to each trial and finding.

CTE01Annotated.csv lists the same candidate type I errors as in CTE01.csv, additionally annotated with the corresponding findings (and notes where applicable) referenced from ClinicalTrials.gov.

Supplementary Information - Nonparametric

Bayesian hypothesis tests using nonparametric statistics. The **R** Markdown knits into an HTML webpage that summarizes the literature and presents simulations for the Wilcoxon signed-rank test, Mann–Whitney U -test, and Kruskal–Wallis H -test. Boxplots compare the $eJAB_{01}$ and Bayes factors constructed from nonparametric statistics (BFNP) across different effect sizes and sample sizes.

Supplementary Information - SI

Preprocessing to address misreported p -values, sample sizes, and test labels.

Supplementary Figures S1–6 (Figures 5–10).

Theorem

Regularity conditions for consistency of $eJAB$ in 11 experiments: two-sample t -test, simple linear regression, simple (binomial) logistic regression, one-way ANOVA, one-way repeated-measures ANOVA, chi-squared test for independence, Cox proportional-hazards regression, conditional logistic regression, Wilcoxon signed-rank test, Mann–Whitney U -test, and Kruskal–Wallis H -test.

`regularity_conditions.RData` contains saved data and functions. The **R** Markdown knits into an HTML webpage.